

Name:

Maths Teacher:

SYDNEY TECHNICAL HIGH SCHOOL



Year 12

Mathematics Extension 2

TRIAL HSC

2016

Time allowed: 3 hours plus 5 minutes reading time

General Instructions:

- Reading time – 5 minutes
- Working time – 3 hours
- Marks for each question are indicated on the question.
- Board - Approved calculators may be used
- In Questions 11- 16, show relevant mathematical reasoning and/or calculations.
- Full marks may not be awarded for careless work or illegible writing
- ***Begin each question on a new page***
- Write using black pen
- All answers are to be in the writing booklet provided
- A BOSTES reference sheet is provided at the rear of this Question Booklet, and may be removed at any time.

Total marks – 100

Section I Multiple Choice

10 Marks

- Attempt Questions 1-10
- Allow 15 minutes for this section

Section II

90 Marks

- Attempt Questions 11-16
- Allow 2 hours 45 minutes for this section

Section I

10 marks

Attempt Questions 1- 10

Allow about 15 minutes for this section

Use the multiple- choice answer sheet located in your answer booklet for Questions 1 -10

1. Which conic has eccentricity $\frac{\sqrt{3}}{3}$?

(A) $\frac{x^2}{3} + \frac{y^2}{2} = 1$

(B) $\frac{x^2}{3^2} + \frac{y^2}{2^2} = 1$

(C) $\frac{x^2}{3} - \frac{y^2}{2} = 1$

(D) $\frac{x^2}{3^2} - \frac{y^2}{2^2} = 1$

2. What value of z satisfies; $z^2 = 20i - 21$?

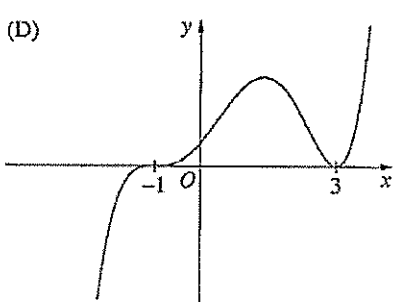
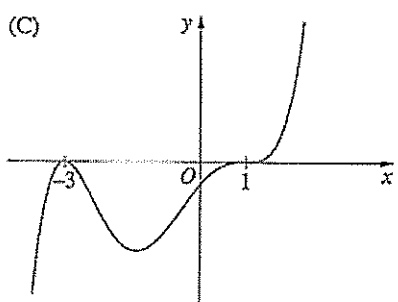
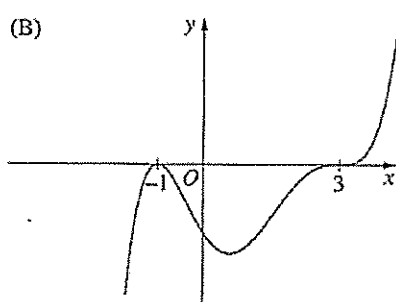
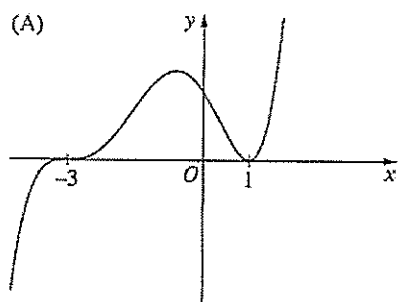
(A) $-2 + 5i$

(B) $2 - 5i$

(C) $2 + 5i$

(D) $5 - 2i$

3. Which graph represents the curve, $y = (x+3)^2(x-1)^3$?



4. The polynomial $2x^4 - 17x^3 + 45x^2 - 27x - 27$ has a triple root at $x = \alpha$.

What is the value of α ?

(A) $-\frac{1}{2}$

(B) $\frac{1}{2}$

(C) -3

(D) 3

5. If $z_1 = 1 + 2i$ and $z_2 = 3 - i$ then $z_1 + \overline{z_2}$ is,

(A) $\frac{1}{2} - \frac{1}{2}i$

(B) $\frac{1}{2} + \frac{1}{2}i$

(C) $4 + 3i$

(D) $\frac{5}{8} + \frac{5}{8}i$

6. Which expression is equal to, $\int \frac{x^2}{\cos^2 x} dx$?

(A) $2x \tan x - 2 \int \tan x dx$

(B) $\frac{1}{3}(x^3 \sec^2 x - \int x^3 \tan x dx)$

(C) $x^2 \tan x - 2 \int x \tan x dx$

(D) $x^2 \tan x - 2 \int x \sec^2 x dx$

7. What is the natural domain of the function $f(x) = \frac{1}{2}(x\sqrt{x^2-1} - \ln(x + \sqrt{x^2-1}))$?

(A) $x \leq -1$ or $x \geq 1$

(B) $-1 \leq x \leq 1$

(C) $x \geq 1$

(D) $x \leq -1$

8. If α, β, δ are the roots of $x^3 + x - 1 = 0$, then an equation with roots

$$\frac{(\alpha+1)}{2}, \frac{(\beta+1)}{2}, \frac{(\delta+1)}{2} \text{ is?}$$

- (A) $x^3 - 3x^2 + 4x - 3 = 0$
- (B) $x^3 + 3x^2 + 4x + 1 = 0$
- (C) $x^3 - 6x^2 + 16x - 24 = 0$
- (D) $8x^3 - 12x^2 + 8x - 3 = 0$

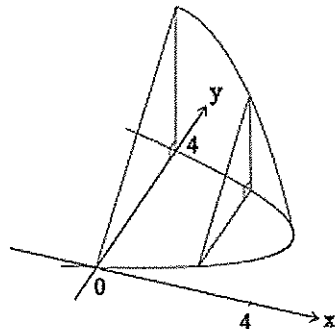
9. The complex number Z satisfies $|Z+2|=1$

What is the smallest positive value of the $\arg(z)$ on the Argand diagram?

- (A) $\frac{\pi}{3}$
- (B) $\frac{5\pi}{6}$
- (C) $\frac{2\pi}{3}$
- (D) $\frac{\pi}{6}$

10. The base of a solid is the region bounded by the parabola $x = 4y - y^2$ and the y axis.

Vertical cross sections are right angled isosceles triangles perpendicular to the x -axis as shown.



Which integral represents the volume of this solid?

- (A) $\int_0^4 2\sqrt{4-x} dx$
- (B) $\int_0^4 \pi(4-x) dx$
- (C) $\int_0^4 (8-2x) dx$
- (D) $\int_0^4 (16-4x) dx$

Question 11 (15 marks)

(a) Express $\frac{18+4i}{3-i}$ in the form, $x+iy$, where x and y are real.

2

(b) Consider the complex numbers $z = -1 + \sqrt{3}i$ and $w = \sqrt{2} \left(\cos\left(\frac{-\pi}{4}\right) + i \sin\left(\frac{-\pi}{4}\right) \right)$

(I) Evaluate $|z|$

1

(II) Evaluate $\arg(z)$

1

(III) Find the argument of $\frac{w}{z}$

2

(c) (i) Find A, B and C such that

3

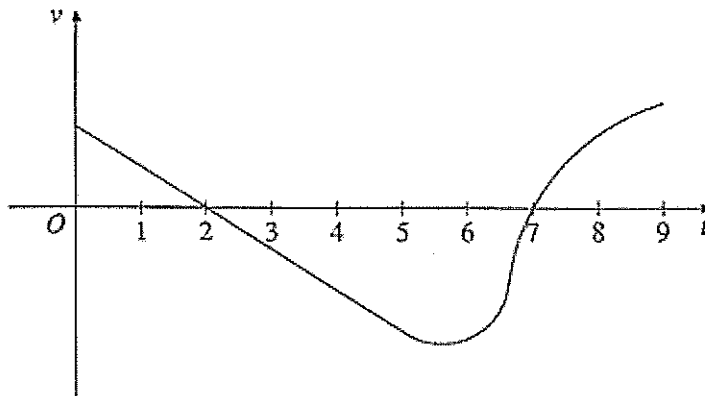
$$\frac{1}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$$

(ii) Hence, or otherwise, find;

$$\int \frac{dx}{x(x^2+4)}$$

2

(d)



A particle moves along the x - axis. At time, $t=0$, the particle is at $x=0$.

Its velocity v at time t is shown on the graph above.

Copy or trace this graph onto your answer page.

(i) At what time is the acceleration the greatest? Explain your answer.

1

(ii) At what time does the particle first return to $x=0$? Explain your answer.

1

(iii) Sketch the displacement time graph for the particle in the interval, $0 \leq t \leq 9$.

2

Question 12 (15 marks) START THIS QUESTION ON A NEW PAGE.

(a) Find $\int x\sqrt{x+1}dx$ 2

(b) Evaluate

(i) $\int_0^{\frac{\pi}{4}} \sin x \cos 2x dx$ 2

(ii) $\int_1^e \frac{\ln x}{x^2} dx$ 2

(c) Find the equation of the normal to the curve, $3x^2y^3 + 4xy^2 = 6 + y$ at the point $(1,1)$. 4

(d)

(i) Prove that,

$$\cos(A-B)x - \cos(A+B)x = 2 \sin Ax \sin Bx$$
 1

(ii) Using the above result, express the equation $\sin 3x \sin x = 2 \cos 2x + 1$,
as a quadratic equation in terms of $\cos 2x$ 2

(iii) Hence, solve, $\sin 3x \sin x = 2 \cos 2x + 1$ for $0 \leq x \leq 2\pi$ 2

Question 13 (15 marks) START THIS QUESTION ON A NEW PAGE.

- (a) The function $y = f(x)$ is defined by the equation;

$$f(x) = \frac{x(x-4)}{4}$$

Without the use of calculus, draw sketches of each of the following, clearly labelling any Intercepts, asymptotes and turning points.

- | | | |
|-------|------------------------|---|
| (i) | $y = f(x)$ | 1 |
| (ii) | $y^2 = f(x)$ | 2 |
| (iii) | $y = \frac{x x-4 }{4}$ | 2 |
| (iv) | $y = \tan^{-1} f(x)$ | 2 |
| (v) | $y = e^{f(x)}$ | 2 |

- (b) Sketch the locus of z satisfying

- | | | |
|------|-----------------------------------|---|
| (i) | $Re(z) = z $ | 2 |
| (ii) | $Im(z) \geq 2$ and $ z-1 \leq 2$ | 2 |

- | | | |
|-----|---|---|
| (c) | Write down the domain and range of $y = 2 \sin^{-1} \sqrt{1-x^2}$ | 2 |
|-----|---|---|

Question 14 (15 marks) START THIS QUESTION ON A NEW PAGE.

- (a) Use the substitution $t = \tan \frac{x}{2}$ to find

4

$$\int_0^{\pi/2} \frac{1}{5 + 4\cos x + 3\sin x} dx$$

- (b) The area enclosed by the curves $y = \sqrt{x}$ and $y = x^2$ is rotated about the y – axis.

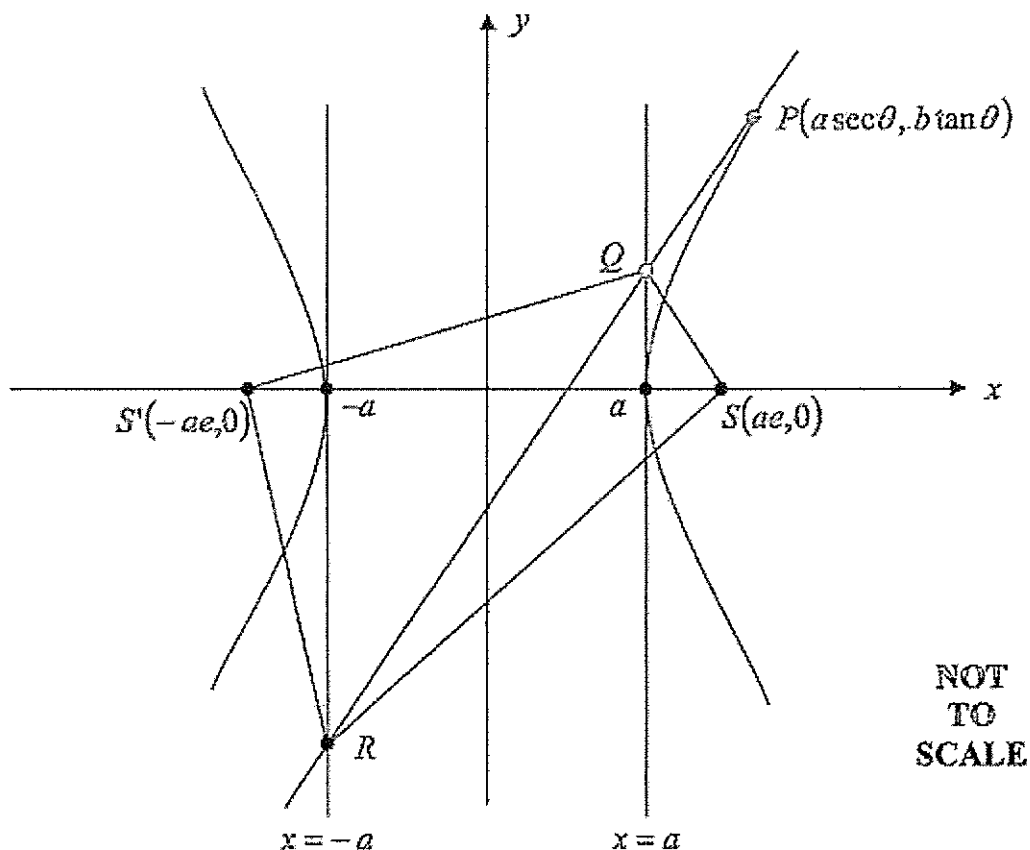
Use the method of *cylindrical shells* to find the volume of the solid formed.

4

Question 14 continues on the next page....

Question 14 continued....

(c)



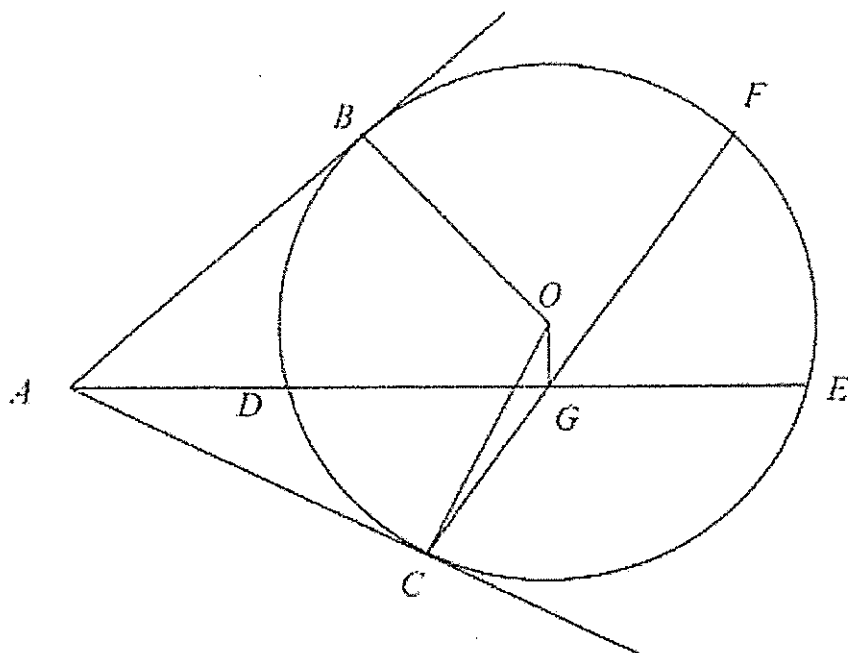
$P(a \sec \theta, b \tan \theta)$ lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

The tangent at P meets the line $x = -a$ and $x = a$ at R and Q respectively.

- | | | |
|-------|---|---|
| (i) | Show that the equation of the tangent is given by $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$. | 2 |
| (ii) | Find the coordinates of Q and R . | 1 |
| (iii) | Show that QR subtends a right angle at the focus $S(ae, 0)$. | 2 |
| (iv) | Deduce that Q, S, R, S' are concyclic. | 2 |

Question 15 (15 marks) START THIS QUESTION ON A NEW PAGE.

- (a) In the diagram, AB and AC are tangents from A to the circle with centre O , meeting the circle at B and C respectively. ADE is a secant of the circle. G is the midpoint of DE . CG produced meets the circle at F .



- | | | |
|-------|--|---|
| (i) | Copy the diagram, using about one third of the page, into your answer booklet and prove that $ABOC$ and $AOGC$ are cyclic quadrilaterals | 3 |
| (ii) | Explain why $\angle OGF = \angle OAC$. | 1 |
| (iii) | Prove that $BF \parallel AE$ | 3 |

(b)

(i) Let $I_n = \int_0^1 x^n \sqrt{1-x^3} dx$ for $n \geq 2$.

Show that: $I_n = \frac{2n-4}{2n+5} I_{n-3}$ for $n \geq 5$ 3

(ii) Hence find I_8 2

- (c) A sequence of numbers is given by $T_1 = 6$ $T_2 = 27$ and $T_n = 6T_{n-1} - 9T_{n-2}$ for $n \geq 3$.

Prove by Mathematical Induction that:

$$T_n = (n+1) \times 3^n \text{ for } n \geq 1$$
3

Question 16 (15 marks) START THIS QUESTION ON A NEW PAGE.

(a) Show that the minimum value of $ae^{mx} + be^{-mx}$ is $2\sqrt{ab}$

if a, b and m are all positive constants.

4

(b) A particle of mass 1 kilogram is projected upwards under gravity (g) with a speed of $2k$

in a medium in which resistance to motion is $\frac{g}{k^2}$ times the square of the speed, where k

is a positive constant.

(i) Show that the maximum height (H) reached by the particle is

$$H = \frac{k^2}{2g} \ln 5$$

3

(ii) Show that the speed with which the particle returns to its starting point

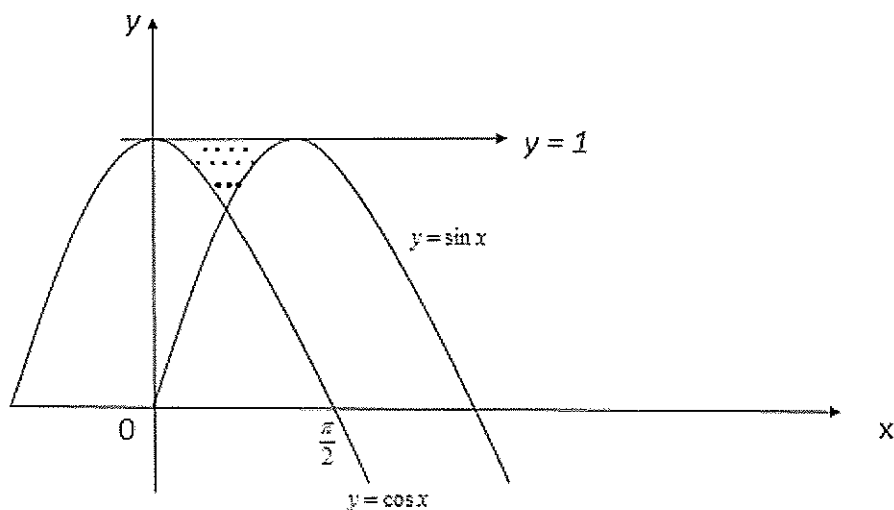
$$\text{is given by } V = \frac{2k}{\sqrt{5}}$$

4

(c) The shaded region in the diagram is bounded by the curves $y = \sin x$, $y = \cos x$

and the line $y = 1$.

This region is rotated around the y – axis.



Calculate the volume of the solid formed, using the process of **Volume by Slicing**.

4

STHS - Ext 2 Trial - Suggestion Solution

Section 1

1. A 2. C 3. C 4. D 5. B
6. C* 7. C 8. D 9. B 10. C
(all given)

Section 2

Question 11

$$a) \frac{18+4i}{3-i} \times \frac{3+i}{3+i}$$

$$= \frac{54 + 18i + 12i - 4}{10}$$

$$= \frac{50 + 30i}{10}$$

$$= 5 + 3i$$

$$b) \omega = \sqrt{2} \text{cis}(-\pi/4) \quad z = -1 + \sqrt{3}i$$

$$i) |z| = \sqrt{1+3}$$

$$= 2$$

$$ii) \arg(z) = 2\pi/3$$

$$iii) \arg\left(\frac{\omega}{z}\right) = \arg(\omega) - \arg(z)$$

$$= -\pi/4 - 2\pi/3$$

$$= -\frac{11\pi}{12}$$

$$c) \frac{1}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$$

$$\therefore 1 = A(x^2+4) + x(Bx+C)$$

$$\text{let } x=0$$

$$1 = A(4) \rightarrow A = 1/4$$

$$\text{equating: } 0 = A + B$$

$$B = -1/4$$

$$0 = C$$

$$\text{ie } A = 1/4 \quad B = -1/4 \quad C = 0$$

ii.

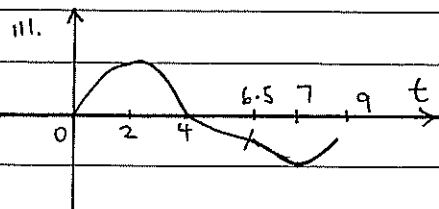
$$\text{Now } \int \frac{dx}{x(x^2+4)} = \int \left(\frac{1}{4x} - \frac{1/4x}{x^2+4} \right) dx$$

$$= \frac{1}{4} \ln x - \frac{1}{8} \ln(x^2+4) + C$$

d) i. $t = 6.5$ (point of inflection on vel. curve is greatest acc)

ii. When the area above the t -axis equals area below $\therefore$ at $t = 4$

x



Question 12

$$ii. \int_1^e \frac{\ln x}{x^2} dx$$

$$a) \int x \sqrt{x+1} dx$$

one method:

$$\text{let } u = x+1$$

$$\frac{du}{dx} = 1 \quad \therefore du = dx$$

$$= \int (u-1) \sqrt{u} du$$

$$= \int u \sqrt{u} - \sqrt{u} du$$

$$= \int u^{3/2} - u^{1/2} du$$

$$=$$

$$\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2}$$

$$= \frac{2}{5} (x+1)^{5/2} - \frac{2}{3} (x+1)^{3/2} + C$$

$$b) i. \int_0^{\pi/4} \sin x \cos 2x dx$$

$$= \int_0^{\pi/4} \sin x (2\cos^2 x - 1) dx$$

$$= \int_0^{\pi/4} 2\sin x (\cos x)^2 - \sin x dx$$

$$= \left[-\frac{2}{3} \cos^3 x + \cos x \right]_0^{\pi/4}$$

$$= -\frac{2}{3} \left(\frac{1}{\sqrt{2}} \right)^3 + \frac{1}{\sqrt{2}} - \left[-\frac{2}{3} (1)^3 + 1 \right]$$

$$= \frac{2}{3\sqrt{2}} - \frac{1}{3}$$

$$= \ln x \cdot \frac{x^{-1}}{-1} - \int \frac{1}{x} \cdot \frac{-1}{x} dx$$

$$= -\frac{\ln x}{x} \Big|_1^e + \int_1^e x^{-2} dx$$

$$= -\left[\frac{1}{e} - 0 \right] + \left[-\frac{1}{x} \right]_1^e$$

$$= -1/e + \left[-1/e - (-1) \right]$$

$$= 1 - 2/e$$

c)

$$3x^2y^3 + 4xy^2 = 6 + y \text{ at } (1,1)$$

$$3x^2 \left[3y^2 \frac{dy}{dx} \right] + y^3 \cdot 6x + 4x \cdot 2y \frac{dy}{dx} + 4y^2 = \frac{dy}{dx}$$

$$6xy^3 + 4x^2 = \frac{dy}{dx} (1 - 9x^2y^2 - 8xy)$$

at (1,1)

$$\frac{dy}{dx} = \frac{10}{-16}$$

$$M_T = -5/8 \quad \therefore M_N = 8/5$$

$$y-1 = 8/5(x-1)$$

$$5y - 5 = 8x - 8$$

$$8x - 5y - 3 = 0$$

Question 12 - con't.

d)

1. Prove that

$$\cos(A-B)x - \cos(A+B)x = 2\sin A x \sin B x$$

$$\text{LHS} = \cos A x \cos B x + \sin A x \sin B x - [\cos A x \cos B x - \sin A x \sin B x]$$

$$= 2\sin A x \sin B x$$

$$= \text{RHS.}$$

II. $\sin 3x \sin x = 2\cos 2x + 1$

$$A=3$$

$$B=1$$

$$\therefore [\cos(3-1)x - \cos(3+1)x] \div 2 = \cos 2x + 1$$

$$\cos 2x - \cos 4x = 2\cos 2x + 2$$

$$\cos 2x - [2\cos^2 2x - 1] = 2\cos 2x + 2$$

$$\cos 2x - 2\cos^2 2x + 1 = 2\cos 2x + 2$$

$$2\cos^2 2x + 3\cos 2x + 1 = 0$$

III. hence,

$$(2\cos 2x + 1)(\cos 2x + 1) = 0$$

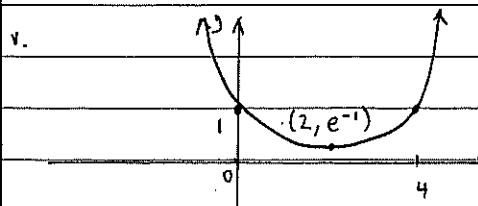
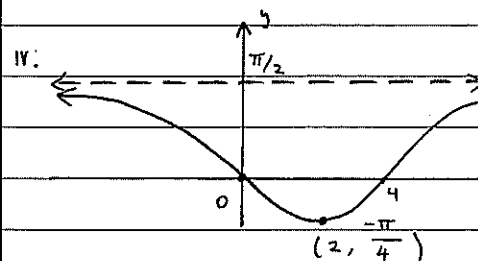
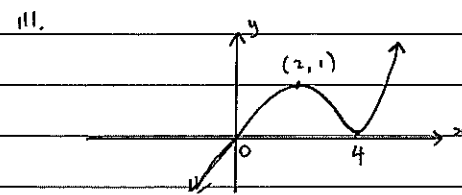
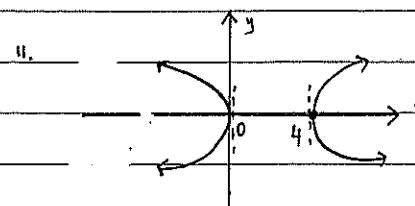
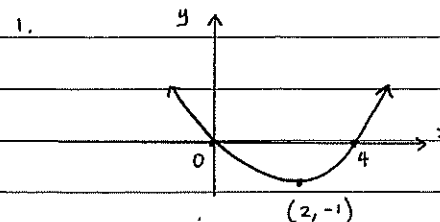
$$\cos 2x = -1/2 \quad \text{or} \quad \cos 2x = -1$$

$$2x = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3} \quad \text{or} \quad 2x = \pi, 3\pi$$

$$x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3} \quad x = \frac{\pi}{2}, \frac{3\pi}{2}$$

Question 13

a) $f(x) = \frac{x(x-4)}{4}$



b) $R(z) = |z|$

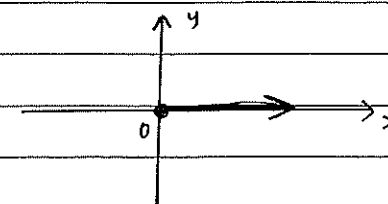
let $z = x + iy$

$$x = \sqrt{x^2 + y^2} \quad x \geq 0$$

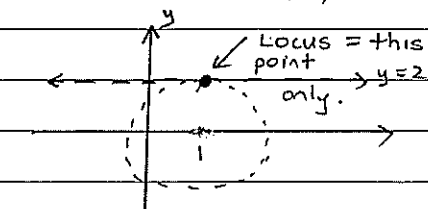
$$x^2 = x^2 + y^2$$

$$y^2 = 0$$

$$y = 0 \quad \text{but} \quad x \geq 0$$



ii. $\text{Im}(z) \geq 2 \quad |z-1| \leq 2$
 $\therefore y \geq 2$ circle centre (1,0)



c) $y = 2\sin^{-1} \sqrt{1-x^2}$

$$y/2 = \sin^{-1} \sqrt{1-x^2}$$

D: $-1 \leq x \leq 1$

R: $0 \leq y \leq \pi$

Question 14.

a) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$x = \pi/2$ $t = \tan \pi/4 = 1$

$x = 0$ $t = \tan 0 = 0$

$\therefore \int_0^1 \frac{1}{5 + 4\left(\frac{1-t^2}{1+t^2}\right) + 3\left(\frac{2t}{1+t^2}\right) \cdot t^2 + 1} \cdot 2 dt$

$= \int_0^1 \frac{2 dt}{5(1+t^2) + 4(1-t^2) + 6t}$

$= \int_0^1 \frac{2}{t^2 + 6t + 9} dt$

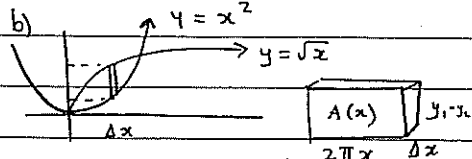
$= 2 \int_0^1 \frac{1}{(t+3)^2} dt$

$= 2 \int_0^1 (t+3)^{-2} dt$

$= \left[\frac{-2}{t+3} \right]_0^1$

$= \left(-\frac{2}{4} \right) - \left(-\frac{2}{3} \right)$

$= \frac{1}{6}$

b) 

$V = \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n 2\pi x (\sqrt{x} - x^2) \Delta x$

$= 2\pi \int_0^1 x^{3/2} - x^3 dx$

$= \left[\frac{2}{5} x^{5/2} - \frac{1}{4} x^4 \right]_0^1 \times 2\pi$

$= 2\pi \left[\frac{2}{5} - \frac{1}{4} - 0 \right] = \frac{3\pi}{10} u$

ii. at Q $x = a$

$\frac{asec\theta}{a} - \frac{\tan\theta y}{b} = 1$

$\frac{1}{\cos\theta} - \frac{y}{b} \frac{\sin\theta}{\cos\theta} = \frac{\cos\theta}{\cos\theta}$

$1 - \frac{y}{b} \sin\theta = \cos\theta$

$y = b(1 - \cos\theta)$

OR $y = b \left(\frac{\sec\theta - 1}{\tan\theta} \right)$

$Q \left[a, \frac{b(1 - \cos\theta)}{\sin\theta} \right]$

(5)

at $R = -a$

$\frac{-asec\theta}{a} - \frac{y \tan\theta}{b} = 1$

$-1 - \frac{y}{b} \sin\theta = \cos\theta$

$y = \frac{-b - b \cos\theta}{\sin\theta}$

OR $y = \frac{-b(1 + \sec\theta)}{\tan\theta}$

iii) $S(ae, 0)$

$m_{SQ} = 0 - \frac{b(1 - \cos\theta)}{\sin\theta}$

$ae - a$

$= \frac{-b(1 - \cos\theta)}{\sin\theta}$

$\frac{a(e-1)}{a(e+1)}$

iv) $m_{as'} = \frac{b(1 - \cos\theta)}{a(e-1)\sin\theta}$

$m_{rs'} = \frac{b(1 + \cos\theta)}{-a(e+1)\sin\theta}$

$\therefore m_{as'} \times m_{rs'}$

$= \frac{b^2(1 - \cos^2\theta)}{-a^2(e^2 - 1)\sin^2\theta}$

$= \frac{b^2}{-b^2}$

$= -1$

$\therefore \angle QSR = 90^\circ$

and $\angle QSR + \angle QSR' = 180^\circ$

making $QSR S'$ a cyclic quad. as opposite angles are supplementary.

$m_{rs} = \frac{b(1 + \cos\theta)}{\sin\theta}$

$\frac{a(e+1)}{a(e+1)}$

Now $m_{sq} \times m_{rs}$

$= \frac{-b(1 - \cos\theta)}{\sin\theta} \times \frac{b(1 + \cos\theta)}{\sin\theta}$

$= \frac{-b^2(1 - \cos^2\theta)}{\sin^2\theta}$

$= \frac{-b^2(e^2 - 1)}{a^2(e^2 - 1)}$

$= -1$ but $a^2(e^2 - 1) = b^2$

$\therefore \angle QSR = 90^\circ$

(6)

Question 15.

Join AO, BF, BC

as $BO = OC$ radii

$\angle OCB = \angle CBO$ (equal angles

$= \alpha$ opposite equal sides)

1. $\angle ABO = \angle OCA = 90^\circ$

(radii to tangent at point of contact is 90°)

$\therefore$ opposite angles in

Now in $\triangle BOC$

$\angle BOC = 180 - 2\alpha$ (angle sum)

$\therefore \angle BFC = 90 - \alpha$

(angle at the circumference is

half the angle at the centre on

arc BC)

$\therefore \angle BFC = \angle FGE$ ($90 - \alpha$)

and the alternate angles

are equal

$\therefore BF \parallel AE$

ABOC are supplementary and

ABOC is a cyclic quadrilateral.

Now, $\angle ABO = 90^\circ$

(AO is a diameter or line from midpt to centre is perpendicular)

$\angle OGA = \angle OCA$ (angles at circumference of circle OGA)

$= 90^\circ$

$\therefore AOGC$ is a cyclic quad

as opposite angles are

supplementary.

ii. $\angle OGF = \angle OAC$

exterior angle of a cyclic

quadrilateral equals opposite

interior angle ($\angle AOGC$).

iii. let $\angle OGF = \angle OAC = \alpha$

$\therefore \angle FGE = 90 - \alpha$ (straight line)

and

$\angle OBC = \angle OAC$ (angles in the same segment of ABOC)

$= \alpha$

Question 15 cont

$$I_n = \int_0^1 x^n \sqrt{1-x^3} dx \quad n \geq 2$$

$$= \int_0^1 x^{n-2} \cdot x^2 \sqrt{1-x^3} dx$$

$$= \left[x^{n-2} (1-x^3)^{3/2} \cdot \frac{-2}{9} \right]_0^1 - \int_0^1 (n-2) x^{n-3} \cdot \frac{-2}{9} (1-x^3) \sqrt{1-x^3} dx$$

$$I_n = 0 + \frac{2(n-2)}{9} \int_0^1 x^{n-3} \sqrt{1-x^3} dx - \frac{2(n-2)}{9} \int_0^1 x^n \sqrt{1-x^3} dx$$

$$I_n \left[1 + \frac{2(n-2)}{9} \right] = \frac{2(n-2)}{9} \int_0^1 x^{n-3} \sqrt{1-x^3} dx$$

$$I_n \left[\frac{9+2n-4}{9} \right] = \frac{2n-4}{9} I_{n-3}$$

$$I_n = \frac{2n-4}{2n+5} I_{n-3}$$

ii)

$$I_8 = \left(\frac{16-4}{16+5} \right) I_5$$

$$= \frac{12}{21} \left[\frac{(10-4)}{10+5} I_2 \right]$$

$$= \frac{12}{21} \cdot \frac{6}{15} \cdot \frac{2}{9}$$

$$= \frac{16}{315}$$

$$\text{Now } I_2 = \int_0^1 x^2 \sqrt{1-x^3} dx$$

$$= \frac{(1-x^3)^{3/2}}{3/2 \cdot -3} \Big|_0^1$$

$$= \left[\frac{-2}{9} (1-x^3)^{3/2} \right]_0^1$$

$$= \frac{-2}{9} \left[0 - 1^{3/2} \right]$$

$$= \frac{2}{9}$$

(7)

(8)

Question 15 con't

c)

$$T_1 = 6 \quad T_2 = 27$$

$$T_n = 6T_{n-1} - 9T_{n-2} \quad n \geq 3$$

$$T_n = (n+1)3^n \quad \text{for } n \geq 1$$

Test $n=1$

$$T_1 = (1+1) \times 3^1$$

$$= 2 \times 3$$

$$= 6 \quad \text{which is given}$$

$\therefore$ True for $n=1$

Assume true for $n=k$

$$\text{ie } T_k = (k+1)3^k$$

where

$$T_k = 6T_{k-1} - 9T_{k-2}$$

Prove true for $n=k+1$

aim to prove

$$T_{k+1} = 6T_k - 9T_{k-1} = (k+1+1)3^{k+1}$$

$$T_{k+1} = 6T_k - 9T_{k-1}$$

$$= 6(k+1)3^k - 9[k(3^{k-1})] \quad \text{By assumption}$$

$$= 2(k+1) \times 3 \times 3^k - 3^2 \cdot k \cdot 3^{k-1}$$

$$= 2(k+1) \cdot 3^{k+1} - k \cdot 3^{k+1}$$

$$= (2k+2-k)3^{k+1}$$

$$= (k+2)3^{k+1}$$

$$= (k+1+1)3^{k+1}$$

as required

(statement Required).

9.

Question 16

$$\text{let } P = ae^{mx} + be^{-mx}$$

$$\text{b) } \begin{array}{c} \uparrow \text{ +ve} \\ \downarrow g \downarrow R \end{array} \quad m = 1kg$$

$$\frac{dP}{dx} = mae^{mx} - mbe^{-mx} = 0$$

$$ae^{mx} = be^{-mx}$$

$$ae^{mx} = \frac{b}{e^{mx}}$$

$$e^{2mx} = b/a$$

$$2mx = \ln(b/a)$$

$$x = \frac{1}{2m} \ln\left(\frac{b}{a}\right)$$

$$1. \quad m\ddot{x} = -mg - \frac{g}{k^2}v^2 \quad m=1$$

$$\therefore \ddot{x} = -g - \frac{g}{k^2}v^2$$

$$v \frac{dv}{dx} = -g \left(\frac{k^2 + v^2}{k^2} \right)$$

$$\frac{dv}{dx} = -g \left(\frac{k^2 + v^2}{v k^2} \right)$$

test

$$\frac{d^2P}{dx^2} = m^2ae^{mx} + m^2be^{-mx}$$

$$\frac{dx}{dv} = -\frac{1}{g} \frac{v k^2}{k^2 + v^2}$$

$$\text{at } x = \frac{1}{2m} \ln(b/a)$$

$$x = -\frac{k^2}{g} \int \frac{v}{k^2 + v^2} dv$$

$$\frac{d^2P}{dx^2} > 0 \quad \text{as } e^{-mx} > 0$$

$$x = -\frac{k^2}{g} \cdot \frac{1}{2} \ln(k^2 + v^2) + C_1$$

and $a, b, m > 0$

$$x=0 \\ v=2k$$

$\therefore$ min value is when

$$x = \frac{1}{2m} \ln(b/a)$$

$$\therefore C_1 = \frac{k^2}{2g} \ln(5k^2)$$

$$P = m \left(\frac{1}{2m} \ln(b/a) \right) - m \left(\frac{1}{2m} \ln(b/a) \right) \therefore$$

$$ae^{mx} + be^{-mx}$$

$$x = -\frac{k^2}{2g} \ln(k^2 + v^2) + \frac{k^2}{2g} \ln(5k^2)$$

$$= ae^{\frac{1}{2} \ln(b/a)} + be^{-\frac{1}{2} \ln(b/a)}$$

$$= ae^{\ln \sqrt{b/a}} + be^{\ln \sqrt{a/b}}$$

max height $v=0$

$$= a \sqrt{\frac{b}{a}} + b \sqrt{\frac{a}{b}}$$

$$x = -\frac{k^2}{2g} \ln k^2 + \frac{k^2}{2g} \ln(5k^2)$$

$$= \frac{k^2}{2g} \ln \left[\frac{5k^2}{k^2} \right]$$

$$= \sqrt{\frac{a^2 b}{a}} + \sqrt{\frac{b^2 a}{b}}$$

$$= \frac{k^2}{2g} \ln 5$$

$$= \sqrt{ab} + \sqrt{ba} = 2\sqrt{ab}$$

10.

Question 16 con't

b) $x=0 \quad t=0 \quad v=0$

$\downarrow + \downarrow mg$

$\uparrow R$

$(m=1)$

$\ddot{x} = g - R$

$\ddot{x} = g - \frac{g}{K^2} v^2$

$v \frac{dv}{dx} = g - \frac{g}{K^2} v^2$

$\frac{dv}{dx} = \frac{g}{v} - \frac{g}{K^2} v$

$= \frac{gK^2 - gv^2}{vK^2}$

$\frac{dx}{dv} = \frac{vK^2}{gK^2 - gv^2}$

$x = \int \frac{vK^2}{gK^2 - gv^2} dv$

$x = \frac{K^2}{g} x - \frac{1}{2} \ln(K^2 - v^2) + C_2$
 $x=0$
 $v=0$

$C_2 = \frac{K^2}{2g} \ln K^2$

$x = \frac{-K^2}{2g} \ln(K^2 - v^2) + \frac{K^2}{2g} \ln K^2$

Now $x = \frac{K^2}{2g} \ln 5$

$\frac{K^2}{2g} \ln 5 = -\frac{K^2}{2g} \ln(K^2 - v^2) + \frac{K^2}{2g} \ln K^2$

$\ln 5 = -\ln(K^2 - v^2) + \ln K^2$

$\ln 5 = \ln \left(\frac{K^2}{K^2 - v^2} \right)$

$5(K^2 - v^2) = K^2$

$-5v^2 = -4K^2$

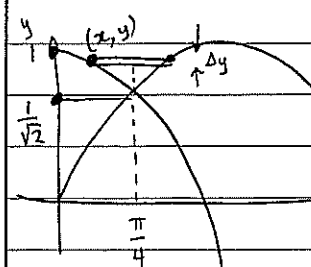
$v^2 = \frac{4K^2}{5}$

$\therefore v = \sqrt{\frac{4K^2}{5}}$

$v > 0$

$v = \frac{2K}{\sqrt{5}}$

Question 16 con't

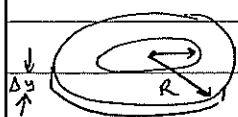


x lies of

$y = \cos x$

$\therefore x = \cos^{-1} y$

(I used symmetry can do $\sin^{-1} y$ & $\cos^{-1} y$)



$r = x = \cos^{-1} y$

$R = \frac{\pi}{2} - x = \frac{\pi}{2} - \cos^{-1} y$

$V = \frac{\pi^2}{2} \left[\frac{\pi}{2} - 2(0) + 0 - \left(\frac{\pi \cdot 1}{2 \sqrt{2}} - \frac{2 \cdot \pi}{\sqrt{2} \cdot 4} + \frac{2}{\sqrt{2}} \right) \right]$

$= \frac{\pi^2}{2} \left[\frac{\pi}{2} - \left(\frac{\pi}{2\sqrt{2}} - \frac{\pi}{2\sqrt{2}} + \frac{2}{\sqrt{2}} \right) \right]$

$= \frac{\pi^2}{2} \left[\frac{\pi}{2} - \sqrt{2} \right] u^3$

$\Delta V = \pi (R^2 - r^2) \Delta y$

$= \pi \left[\frac{\pi}{2} - \cos^{-1} y - \cos^{-1} y \right] \left[\frac{\pi}{2} - \cos^{-1} y + \cos^{-1} y \right] \Delta y$

other solutions

such as

$\frac{\pi^2}{2} \int_{1/\sqrt{2}}^1 2 \sin^{-1} y - \frac{\pi}{2} dy$

can be used.

$= \pi \left[\frac{\pi}{2} - 2 \cos^{-1} y \right] \left[\frac{\pi}{2} \right] \Delta y$

$= \frac{\pi^2}{2} \left[\frac{\pi}{2} - 2 \cos^{-1} y \right] \Delta y$

Total volume

$= \lim_{\Delta y \rightarrow 0} \sum_{1/\sqrt{2}}^1 \frac{\pi^2}{2} \left[\frac{\pi}{2} - 2 \cos^{-1} y \right] \Delta y$

$= \frac{\pi^2}{2} \int_{1/\sqrt{2}}^1 \left[\frac{\pi}{2} - 2 \cos^{-1} y \right] dy$

$= \frac{\pi^2}{2} \left[\frac{\pi}{2} y - \left[2y \cos^{-1} y - \int 2y \cdot \frac{-1}{\sqrt{1-y^2}} dy \right] \right]$

$= \frac{\pi^2}{2} \left[\frac{\pi}{2} y - 2y \cos^{-1} y + 2\sqrt{1-y^2} \right]_{1/\sqrt{2}}^1$

(11)

(12)